Belief representations in RL agents

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1 Introduction & assumptions

In the table below, we explain the notation used in the experiments and the rest of the document.

$s \in \mathcal{S}$, M	memory state of a finite-state controller; $M = \lvert\mathcal{S}\rvert$
a_t, a_t^i, a_t^{-i}	joint move of round t; agent i's entry; the others' entries
$\pi_\theta(a \mid s)$	action table: move distribution of state s
$\nu_\theta(\tilde{s} \mid s, a)$	memory update: next-state distribution after joint move a
$\theta = (\theta^\pi, \theta^\nu)$	learned logits of both tables, frozen before any probing
α	entropy coefficient of the RL objective
K, k	# behaviorally distinct frozen policies; which one plays this game (hidden)
b_t	exact posterior over (k, s_t) given public history (the filter)
$H(s_t \mid \text{history})$	hiddenness: entropy of the exact belief
R^2, L	probe fit against the filter; window length of the short-memory baselines

Necessary conditions to have a belief geometry?

1. A ground truth system with hidden states (this matters for calculating the representation faithfulness)
2. Nondegenerate belief over the hidden states (this matters to have a nontrivial belief geometry). Or is it the states themselves as nondegenerate?
3. The belief must not be recoverable by short memory (slow mixing).
4. An incentive to represent beliefs: Representing hidden states must in some way serve the agent.
5. Anything else missing?

While reinforcement learning does not necessarily create hidden states, they are still achievable: the belief over s_t is nondegenerate exactly when the public history does not determine s_t. One caution on terms: the belief transducer is always unifilar in the sense of [2] Proposition 2 (the next belief is a deterministic function of the current belief and the round's tokens); degeneracy is the separate property that the belief itself is a point mass. If s_0 is known and ν_θ is deterministic given (s, a), then s_t is computable from the joint moves and the belief is degenerate for all t. Three mechanisms produce nondegeneracy:

1. Stochastic memory updates: $\nu_\theta(\cdot \mid s, a)$ has entropy.
2. Private personas: which member of a trained population is playing, sampled per game and frozen.
3. Private observation channels: the agent reads the joint move through a noisy channel, which leaves the realized update uncertain to the observer.

Important note: Policy-gradient methods driven to convergence approach deterministic policies, and a greedy π with a deterministic ν kills mechanism 1 and (via behavioral distinguishability collapsing to a lookup) most of the value of mechanism 2, making entropy regularization crucial. The pre-training checks below measure the consequence directly, as $H(s_t \mid \text{history})$ under the exact filter, and the pre-registered expectation is that this quantity falls toward zero as the entropy coefficient does.

2 Experiment A

Instead of probing the agents that play the game, we can keep the actor and the observer separate: RL agents play the IPD/cooperateive games, a separate observer transformer watches their games, and we probe the observer. The players are finite-state controllers (FSCs) trained by policy gradient and then frozen into a population of K behaviorally distinct personas, so the hidden states we probe for are produced by optimization rather than written by hand. The observer is trained by next-token prediction on the transcripts and does not act. For prediction, the belief state is provably the right summary of the past, which makes learning the representations instrumentally useful for the observer.[1] The probe target is the exact Bayesian filter b_t over the pair (persona k, memory state s_t). The experiment setup is as follows:

1. The agents are trained by policy gradient on their own return. Each agent is a finite state controller with M memory states and two softmax tables, π_θ for the moves and

[1] In experiment B the same incentive must instead be designed into the environment.

ν_θ for the memory updates [7]. In addition to self play, we add in some agents who do well in the IPD (e.g. tit-for-tat, GRIM, Pavlov from the traditional IPD tournaments) to facilitate diversity in states. These library agents shape what is learned, but they do not have to enter the frozen population

2. We minimize each frozen machine, deduplicate by isomorphism of the minimized machines, and keep the remaining K distinct ones. Before training any observer, we run the pre-training checks against the exact filter of the frozen population (train the observer only if all pass):

   (i) hiddenness: $H(s_t \mid \text{history}) > 0$ sustained late in the game;

   (ii) degeneracy diagnostics: fraction of near-deterministic rows of π and ν, and the collapse checks of the host game (for the public-goods game, early-versus-late mean contribution).

3. After freezing the population, the observer transformer is trained by next-token prediction on its games, then frozen as well. Linear probes are fit from the ground-truth latents (the exact filter b_t over (k, s_t)) to the observer's residual stream. The predicted geometry of the probe target is the mixed-state presentation: belief trajectories on the simplex over hidden states. Probe R^2 is compared against window baselines, the untrained model, and the exact-filter ceiling.

Our challenges/goals are as follows:

1. Producing hiddenness through optimization. Policy gradient driven to convergence approaches deterministic tables, and a deterministic ν with a known s_0 makes s_t computable from the public moves, so we must avoid collapsing to a deterministic policy.

   > More concretely: track $H(s_t \mid \text{history})$ under the exact filter late in the game, across the entropy-coefficient sweep. The pre-registered expectation is that hiddenness falls toward zero as α does; the usable populations are the ones where it stays bounded away from zero.

2. Producing diversity through optimization ($K \geq 2$ after deduplication), and stating the claim at the strength we actually earned. Do all seeds land on the same equilibrium?

   > More concretely, a ladder of fallback claims: (a) pure self-play gives $K \geq 2$ (the strongest claim); (b) $K \geq 2$ appears somewhere in a spread over α, M, horizon and freeze point

3 Experiment B

Instead of having a separate observer agent, we can use transformer RL agents who are both the observer and the actor. Moreover, interacting with space potentially could allow us to investigate richer cooperative behaviors. The following is an experiment setup with the suggestions from PIBBSS fellow Jonathan Elsworth Eicher from his previous work on Arnold tongues. We suggest modeling cooperative scenarios through multiagent interactions with periodic reward sources in a gridworld. Reward objects (hearts) are produced by converters that switch on and off as square waves with period T; an agent in a hallway of length d_h must phase-lock its movement to the converter cycle, which is possible only above the critical period $T_c = d_h/r_a$. Figure 1 illustrates the three-stage pipeline:

1. A transformer is pretrained by next-token prediction on games played by a population of K scripted personas, each a (goal, value, memory) slot configuration to establish a representational circuitry

2. RL post-training on the multi-agent map, whose converters emit hearts as square waves $H_i(t)$ with periods T_i. In the simplest example, this is one agent and one reward source (converter). Agents are trained by policy gradient on return, $+R_H$ per heart collected and $-R_S$ per step. How could we encourage the agents to represent the other agent? I hope with enough pressure, an agent might want to jeopardize the other agent's reward source to get more hearts for itself or alternatively they fall into equilibria, eg. agent i only goes to converter j and agent k only goes to converter l, even though they are both capable of going to both of them within one period. Currently having a hard time consolidating this type of behavior with the circuitry discussed at 1.

3. After freezing the agents, linear probes are fit from the ground-truth latents (the exact Bayesian filter over persona and phase) to the acting agent's residual stream. The predicted geometry of the probe target is the mixed-state presentation: belief trajectories on the simplex over hidden states. Probe R^2 is compared against window baselines [2], the untrained model, and the exact-filter ceiling.

Some analogous objects are as follows:

- Hidden states of the agents are their policies from pretraining
- Hidden states of reward sources are the phase of the converter cycle. This is the clock position $t \bmod T$ of the converter, which is hidden from the agent unless it can see the converter state directly.[3]

[2] In order to test whether representing the converter/other agents is beneficial; if a short window linear probe or other simple program is enough to ?? a bit of a confusion here. enough for what? enough to maximize reward? if it it is enough, maintaining a belief simplex is distinctly redundant.

[3] With the adaptive period rule, the period itself becomes a hidden state as well, making the converter input-driven by the agent's successes and failures.

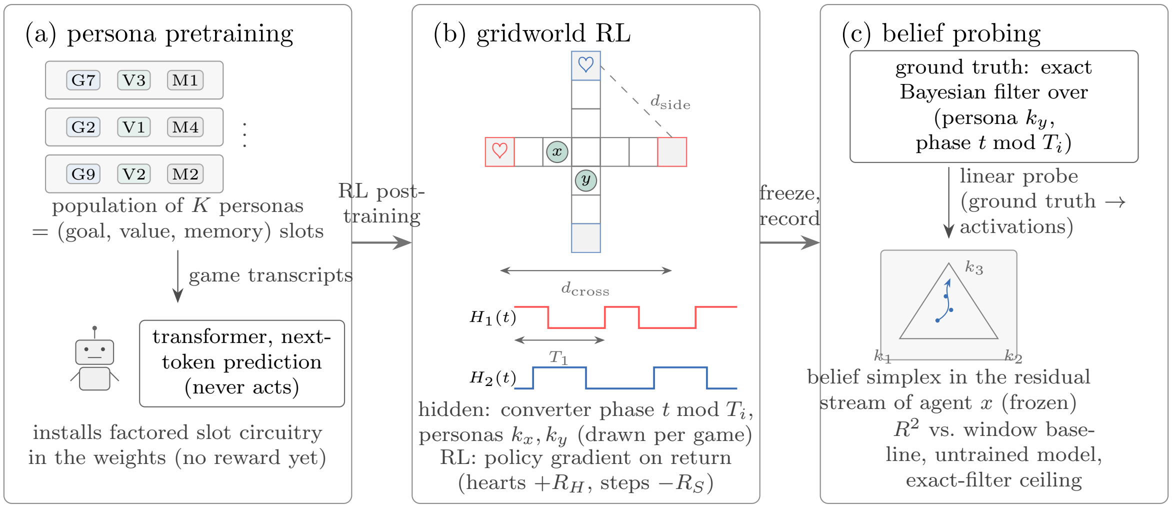

Figure 1: Experiment setup.

- Observability: which parts of the world any agent k_i sees

Our challenges/goals are as follows:

1. Representing the periodic reward sources as a transducer with hidden memory, where the hidden state is the phase of the converter cycle. The agent must learn to infer this hidden state from its observations and actions.

2. After pretraining (which includes the right configuration of agent structure with goal, value, and memory), the agent can have the right background to be able to learn the policy (hidden states). It is a nontrivial challenge to configure an environment that would facilitate/encourage representing the other agent and we must avoid collapsing to a memoryless policy. The suggestion to have the configuration of agent structure is also to support this, however the capability of the model in representation does not guarantee that it will learn to represent the other agent.

3. With adjusting the ratio of agents/rewards/hallway length, we can explore the emergence of swarm behavior and investigate the change in representation in individual agents. Do representations converge?

   > More concretely: When fitting probes from the ground-truth latents to each agent's activations, do the fitted subspaces across agents and seeds converge (consistently small principal angles)?

   Does a 'swarm representation' look fundamentally different than a preexisting individual strategy?

> More concretely: Probe for per-individual latents (agent j's persona, agent j's phase/position) versus aggregate statistics (how many agents are headed to converter 1, crowding at my target). The mean-field signature would be: as n grows past what the context can track jointly, per-identity R^2 falls while aggregate-statistic R^2 holds.

4 Blockers?

There are two shared failure modes for both experiments. First: all seeds may converge to memoryless or near-deterministic policies, in which case $H(s \mid \text{history}) \approx 0$ and there is no filtering problem; the mitigation is an entropy floor during training, and if hiddenness still dies, self play is unable to produce filterable opponents. Second: learned ν may mix fast, so the beliefs are forgotten quickly and short windows recover it.

Questions

- If all seeds land on one equilibrium ($K = 1$ after deduplication), what can we do? Re-training?
- What can we do to encourage slow mixing?
- Other failure modes?

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